

NANYANG TECHNOLOGICAL UNIVERSITY**SEMESTER 2 EXAMINATION 2024-2025****EE7207 – NEURAL NETWORKS & DEEP LEARNING**

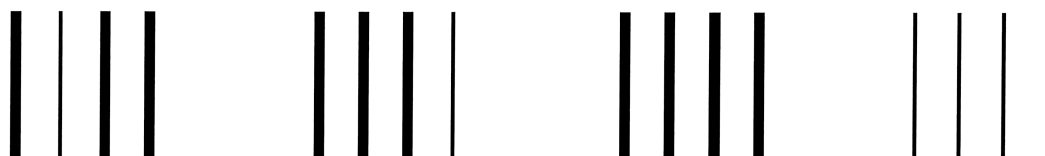
April / May 2025

Time Allowed: 3 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 4 pages.
 2. Answer all 4 questions.
 3. All questions carry equal marks.
 4. This is a closed book examination.
 5. Unless specifically stated, all symbols have their usual meanings.
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1. There are four barcodes as shown in Figure 1:

**Figure 1**

- (a) If a Hopfield neural network is used to store the four barcodes, sketch the architecture of the Hopfield neural network, determine the weight matrix, and check whether the four barcodes could be correctly retrieved by the Hopfield neural network designed (Hint: you can use 1 and -1 to represent the thick and thin lines, respectively).
(10 Marks)
- (b) If a bi-directional associative memory (BAM) neural network is used to store the four barcodes, sketch the architecture of the BAM neural network, determine the weight matrices, and check whether the four barcodes could be correctly retrieved by the BAM neural network designed.
(10 Marks)

Note: Question No. 1 continues on page 2.

- (c) Discuss the limitations of Hopfield neural networks in storing fundamental memories.

(5 Marks)

2. (a) Assume that a kernel support vector machine (SVM) is adopted to solve a two-class classification problem. After training the kernel SVM, four support vectors, $\mathbf{s}_1, \mathbf{s}_2, \mathbf{s}_3, \mathbf{s}_4$, are obtained, and the Lagrange multipliers corresponding to the four support vectors are:

$$\alpha_1 = 0.76, \alpha_2 = 0.31, \alpha_3 = 0.29, \alpha_4 = ?$$

The class labels of the four support vectors are $1, -1, -1, -1$, respectively.

- (i) Calculate the value of α_4 .
 (ii) The discriminant function of the kernel SVM can be expressed as:

$$f(\mathbf{x}) = \sum_{i=1}^4 w_i K(\mathbf{x}, \mathbf{s}_i)$$

where K denotes the kernel function, and $\mathbf{x}$ is the input feature vector. Calculate the values of $w_i, i = 1, 2, 3, 4$.

(8 Marks)

- (b) Assume that a Gaussian radial basis function (RBF) neural network is used to solve the classification problem in part (a). List all the unknown parameters in the Gaussian RBF neural network and describe the two-stage method for the learning of the unknown parameters.

(11 Marks)

- (c) Suggest three ways to address the overfitting problem in machine learning.

(6 Marks)

3. (a) Consider the sentence: "fred fed ted bread". Perform Byte Pair Encoding (BPE) step-by-step. Only perform 2 merges. At each merge:

- (i) Identify the most frequent pair.
 (ii) Merge the pair in all occurrences.
 (iii) Update your vocabulary.

Your answer should clearly demonstrate each intermediate step.

(8 Marks)

Note: Question No. 3 continues on page 3.

- (b) Imagine you want to train a GPT-like assistant. The training process can be broken down into multiple steps.

- (i) List each of these training steps in the correct sequence.
- (ii) For each step, explain the dataset used, the training task, and the output of this step (what model is produced).

(8 Marks)

- (c) You are given the following reference summary (R) and generated summary (H):

Reference (R):

“large language models power modern NLP systems with impressive generative capabilities”

Generated (H):

“modern NLP uses large language models for generative tasks”

Answer the following:

- (i) List all bigrams from each summary (assume tokenization by whitespace).
- (ii) Identify the overlapping bigrams between R and H.
- (iii) Compute ROUGE-2 Recall, ROUGE-2 Precision, and ROUGE-2 F1.

(9 Marks)

4. You are tasked with designing a deep learning model to classify rare cancer subtypes from medical images. The dataset comprises 1,000 annotated images with a significant class imbalance (approximately 10:1). The final model will be deployed on mobile devices. Answer the following questions:

- (a) Propose three effective data augmentation strategies (beyond simple rotations and flips) that are suitable for the task. For each strategy, briefly explain:
 - (i) How it generates more diverse training samples.
 - (ii) Why it is expected to mitigate the effects of limited data and class imbalance.

(6 Marks)

Note: Question No. 4 continues on page 4.

- (b) Consider a convolutional layer that outputs a feature map with dimensions $(N, C, H, W) = (32, 256, 28, 28)$, where 32 is the batch size, 256 is the number of channels, and 28×28 are the spatial dimensions. When applying batch normalization on this convolutional output, what is the shape of the running mean used during training? Select the correct answer from the following choices and provide a concise justification for your selection.

- (i) (256,)
- (ii) (32, 256)
- (iii) (256, 28, 28)
- (iv) (32, 28, 28)

(5 Marks)

- (c) Discuss the role of Batch Normalization in training this network. (**Hint:** Focusing on how it stabilizes and accelerates convergence, especially in scenarios with limited and imbalanced data.)

(6 Marks)

- (d) Assume that MobileNet is adopted as the pretrained model because of the depthwise separable convolution. Describe how the key ideas behind it differ from standard convolutions.

(4 Marks)

- (e) A colleague suggests replacing the convolutional backbone with a Vision Transformer (ViT) to potentially improve classification performance. Evaluate the benefits and drawbacks of using a Vision Transformer compared to a convolutional backbone in terms of performance, data efficiency, and computational cost. Based on your evaluation, what is your opinion on the suggestion of replacing the convolutional backbone with a ViT in this scenario? Justify your answer.

(4 Marks)

END OF PAPER

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EE7207 NEURAL NETWORKS & DEEP LEARNING

Please read the following instructions carefully:

1. **Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**
2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.
3. Please write your Matriculation Number on the front of the answer book.
4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.